

Syntomic Tate Twists over p -adic Local Fields and the Cyclotomic Torsion Groups

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Abstract

Let $K/\mathbb{Q}_p$ be a finite extension with ring of integers $\mathcal{O}_K$. We compute the BMS syntomic complexes

$$\mathbb{Z}_p(i)(\mathcal{O}_K) = \mathrm{R}\Gamma_{\mathrm{syn}}(\mathrm{Spec} \mathcal{O}_K, \mathbb{Z}_p(i))$$

for all integer weights $i \in \mathbb{Z}$. For $i < 0$ the complexes vanish. For $i \geq 0$ the calculation reduces, through the comparison of Bhatt–Mathew with Geisser–Sato–Schneider p -adic étale Tate twists, to local Galois cohomology. The answer is expressed in terms of the completed unit group $\widehat{\mathcal{O}_{Kp}^\times}$, the rank $n_K = [K : \mathbb{Q}_p]$, and the finite cyclotomic torsion groups

$$W_m(K) = H^0(K, \mathbb{Q}_p/\mathbb{Z}_p(m)).$$

We also give a complete calculation of $W_m(K)$ from the image of the p -adic cyclotomic character. For odd p this gives a closed formula in terms of the tame order $[K(\mu_p) : K]$ and the wild cyclotomic level; for $p = 2$ it gives the corresponding formula using the sign and pro-2 projections.

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1 Statement and comparison

Let $K/\mathbb{Q}_p$ be a finite extension, let $\mathcal{O}_K$ be its ring of integers, let k be its residue field, and set

$$n_K = [K : \mathbb{Q}_p].$$

Fix a separable closure $\overline{K}$ of K , and put

$$G_K = \text{Gal}(\overline{K}/K).$$

For any integer m , define

$$W_m(K) := H^0(G_K, \mathbb{Q}_p/\mathbb{Z}_p(m)).$$

For $m \neq 0$ this will be finite by Theorem 4.3; for $m = 0$ it is $\mathbb{Q}_p/\mathbb{Z}_p$. If A is a topological abelian group, write

$$A^\vee = \text{Hom}_{\text{cts}}(A, \mathbb{Q}_p/\mathbb{Z}_p)$$

for its Pontryagin dual, and write

$$\widehat{\mathcal{O}_{Kp}^\times} := \varprojlim_r \mathcal{O}_K^\times / (\mathcal{O}_K^\times)^{p^r}$$

for the p -adic completion of the unit group. We use cohomological grading: $M[-r]$ denotes the complex with M in cohomological degree r .

We use the following convention for syntomic complexes. The only syntomic bases used below are $X = \text{Spec } \mathcal{O}_K$ and $s = \text{Spec } k$; both are quasisyntomic, so the BMS construction applies to them. For such a scheme Y , let

$$\mathbb{Z}_p(i)_Y^{\text{syn}}$$

denote the p -complete syntomic Tate-twist complex on the small étale site of Y , as constructed by Bhatt–Morrow–Scholze; see [3, Theorem 1.12 and Remark 1.14]. Its global sections are written

$$\mathbb{Z}_p(i)(Y) := R\Gamma_{\text{syn}}(Y, \mathbb{Z}_p(i)) := R\Gamma(Y_{\text{ét}}, \mathbb{Z}_p(i)_Y^{\text{syn}}).$$

The convention includes the BMS negative-weight normalization $\mathbb{Z}_p(i)_Y^{\text{syn}} = 0$ for $i < 0$.

Definition 1.1 (Sato twists in the trait case). *Let $n \geq 1$ and $i \geq 0$. Sato’s finite-level p -adic étale Tate twist*

$$\mathfrak{I}_n(i)_Y \in D^b(Y_{\text{ét}}, \mathbb{Z}/p^n\mathbb{Z})$$

is the $\mathbb{Z}/p^n\mathbb{Z}$ -coefficient étale complex constructed in [6, §4.2]. In this article only the cases $Y = X = \text{Spec } \mathcal{O}_K$ and $Y = s = \text{Spec } k$ occur.

For $Y = s$, which has characteristic p , the twist is the logarithmic Hodge–Witt complex

$$\mathfrak{I}_n(i)_s = W_n\Omega_{s,\log}^i[-i], \quad W_n\Omega_{s,\log}^0 = \mathbb{Z}/p^n\mathbb{Z}.$$

Here $W_n\Omega_{s,\log}^i$ denotes the logarithmic de Rham–Witt sheaf in Sato’s convention: for $i \geq 1$ it is generated étale-locally by symbols

$$d\log[a_1] \wedge \cdots \wedge d\log[a_i], \quad a_j \in \mathcal{O}^\times,$$

with $[a_j]$ the Teichmüller lift. This is the convention used by Sato and by the Bloch–Kato description of p -adic vanishing cycles; see [6, §4.2] and [2, Theorem 14.1 and §6.6].

For $Y = X$, let $j : \text{Spec } K \hookrightarrow X$ and $\iota : s \hookrightarrow X$. The weight 0 twist is $\mathfrak{I}_n(0)_X = \mathbb{Z}/p^n\mathbb{Z}_X$. For $i \geq 1$, $\mathfrak{I}_n(i)_X$ is the object fitting into Sato’s defining distinguished triangle

$$\iota_* W_n\Omega_{s,\log}^{i-1}[-i-1] \longrightarrow \mathfrak{I}_n(i)_X \longrightarrow \tau_{\leq i} Rj_* \mu_{p^n}^{\otimes i} \xrightarrow{\sigma} \iota_* W_n\Omega_{s,\log}^{i-1}[-i].$$

The morphism σ is induced on the top cohomology sheaf by Sato's logarithmic residue map from p^n -torsion Kummer symbols on K to logarithmic Hodge–Witt forms on k ; in the notation of [6, Definition 4.2.4] it is the residue entering the mapping-fiber definition of $\mathfrak{I}_n(i)_X$. Equivalently, for $0 \leq q < i$,

$$\mathcal{H}^q(\mathfrak{I}_n(i)_X) \simeq R^q j_* \mu_{p^n}^{\otimes i},$$

while $\mathcal{H}^i(\mathfrak{I}_n(i)_X)$ is the kernel of

$$R^i j_* \mu_{p^n}^{\otimes i} \xrightarrow{\sigma} \iota_* W_n \Omega_{s, \log}^{i-1}.$$

Thus, in the present one-dimensional mixed-characteristic situation, $\mathfrak{I}_n(i)_X$ is the complex that glues the Kummer twist on the generic point to the logarithmic Hodge–Witt twist on the closed point, with the gluing controlled by residue. It is not merely $Rj_* \mu_{p^n}^{\otimes i}$; the fiber of σ removes the wrong top residue and gives the object satisfying Sato's localization, purity, and product properties.

For $Y = X$ or $Y = s$, set

$$\mathbb{Z}_p(i)_Y^{\text{GSS}} := R \varprojlim_n \mathfrak{I}_n(i)_Y.$$

This is the Geisser–Sato–Schneider p -adic etale Tate twist used below. On the generic point it restricts to the rank-one continuous $\mathbb{Z}_p$ -module on which $g \in G_K$ acts by $\chi_K(g)^i$.

Theorem 1.2 (Cohomology of syntomic Tate twists over $\mathcal{O}_K$). *Let*

$$C_i := \mathbb{Z}_p(i)(\mathcal{O}_K) = R\Gamma_{\text{syn}}(\text{Spec } \mathcal{O}_K, \mathbb{Z}_p(i)).$$

Then the following hold.

(a) *If $i < 0$, then $C_i \simeq 0$.*

(b) *If $i = 0$, then*

$$H^0(C_0) \cong \mathbb{Z}_p, \quad H^1(C_0) \cong \mathbb{Z}_p,$$

and $H^r(C_0) = 0$ for $r \neq 0, 1$.

(c) *If $i = 1$, then*

$$H^1(C_1) \cong \widehat{\mathcal{O}_{Kp}^\times},$$

and $H^r(C_1) = 0$ for $r \neq 1$. Noncanonically,

$$\widehat{\mathcal{O}_{Kp}^\times} \cong \mathbb{Z}_p^{n_K} \oplus \mu_{p^\infty}(K).$$

(d) *If $i \geq 2$, then*

$$H^0(C_i) = 0, \quad H^1(C_i) \cong H^1(K, \mathbb{Z}_p(i)), \quad H^2(C_i) \cong W_{1-i}(K)^\vee,$$

and $H^r(C_i) = 0$ for $r \notin \{1, 2\}$. Moreover $H^1(K, \mathbb{Z}_p(i))$ is a finitely generated $\mathbb{Z}_p$ -module of rank n_K , its torsion subgroup is $W_i(K)$, and hence noncanonically

$$H^1(K, \mathbb{Z}_p(i)) \cong \mathbb{Z}_p^{n_K} \oplus W_i(K).$$

Corollary 1.3 (The complexes in $D(\mathbb{Z}_p)$). *In the derived category $D(\mathbb{Z}_p)$, noncanonically,*

$$C_0 \simeq \mathbb{Z}_p \oplus \mathbb{Z}_p[-1], \quad C_1 \simeq \widehat{\mathcal{O}_{Kp}^\times}[-1],$$

and, for $i \geq 2$,

$$C_i \simeq H^1(K, \mathbb{Z}_p(i))[-1] \oplus W_{1-i}(K)^\vee[-2].$$

For $i < 0$, $C_i \simeq 0$.

Remark 1.4. *The splittings in Corollary 1.3 are not canonical. The cohomology groups in Theorem 1.2, and the exact sequences used in the proof, are canonical.*

Lemma 1.5 (Closed-point calculation for Sato's twist). *Let $X = \operatorname{Spec} \mathcal{O}_K$, let $s = \operatorname{Spec} k$, and let*

$$j : \operatorname{Spec} K \hookrightarrow X, \quad \iota : s \hookrightarrow X$$

be the generic and closed immersions. For $n \geq 1$ and $i \geq 1$, Sato's finite-level twist satisfies

$$j^* \mathfrak{I}_n(i)_X \simeq \mu_{p^n}^{\otimes i} \quad \text{and} \quad \iota^! \mathfrak{I}_n(i)_X \simeq \mathfrak{I}_n(i-1)_s[-2].$$

More explicitly, after passing to the étale stalk at a geometric point of the closed point, the second isomorphism is normalized by the logarithmic residue map

$$\{\pi, u_1, \dots, u_{i-1}\} \longmapsto d \log(\bar{u}_1) \wedge \dots \wedge d \log(\bar{u}_{i-1}),$$

with target $W_n \Omega_{\bar{k}, \log}^{i-1}$; for $i = 1$, the empty wedge is $1 \in W_n \Omega_{\bar{k}, \log}^0 = \mathbb{Z}/p^n \mathbb{Z}$.

Proof. This is the strict henselian stalk calculation behind Sato's absolute purity theorem in codimension one. We spell it out because the normalization of the residue is used later.

The generic assertion follows directly from Definition 1.1. Applying j^* to the defining triangle kills the two terms supported on s , and gives

$$j^* \mathfrak{I}_n(i)_X \simeq j^* \tau_{\leq i} Rj_* \mu_{p^n}^{\otimes i} \simeq \mu_{p^n}^{\otimes i}.$$

We next compute $\iota^! \mathfrak{I}_n(i)_X$ from the same defining triangle. The assertion is étale-local on s , so compute at a geometric point $\bar{s} \rightarrow s$. Let R be the henselian discrete valuation ring obtained from this geometric étale stalk; its residue field $\bar{k}$ is separably closed. Let $L = \operatorname{Frac}(R)$, and choose a uniformizer $\pi \in R$. For a complex F on $X_{\text{ét}}$,

$$\mathcal{H}^q(R\iota^! F)_{\bar{s}} = H_{\bar{s}}^q(\operatorname{Spec} R, F),$$

the local cohomology at the closed point.

Put $T = \mathfrak{I}_n(i)_X$. The localization long exact sequence for $(\operatorname{Spec} R, \operatorname{Spec} L)$, together with the generic identification just proved, gives

$$\dots \rightarrow H_{\bar{s}}^q(\operatorname{Spec} R, T) \rightarrow H^q(\operatorname{Spec} R, T) \rightarrow H^q(L, \mu_{p^n}^{\otimes i}) \rightarrow H_{\bar{s}}^{q+1}(\operatorname{Spec} R, T) \rightarrow \dots$$

Since $\operatorname{Spec} R$ has the étale topology of the geometric stalk, the groups $H^q(\operatorname{Spec} R, T)$ are the stalks of the cohomology sheaves of T at $\bar{s}$. By Definition 1.1, the map

$$H^q(\operatorname{Spec} R, T) \longrightarrow H^q(L, \mu_{p^n}^{\otimes i})$$

is an isomorphism for $q < i$, and for $q = i$ it identifies $H^i(\operatorname{Spec} R, T)$ with the kernel of the residue map

$$\partial_R : H^i(L, \mu_{p^n}^{\otimes i}) \longrightarrow W_n \Omega_{\bar{k}, \log}^{i-1}.$$

This ∂_R is the Bloch–Kato residue appearing in Sato's definition. It is surjective by the symbol description of logarithmic Hodge–Witt sheaves: after lifting elements of $\bar{k}^\times$ to units of R ,

$$\partial_R \{\pi, u_1, \dots, u_{i-1}\} = d \log(\bar{u}_1) \wedge \dots \wedge d \log(\bar{u}_{i-1}) \in W_n \Omega_{\bar{k}, \log}^{i-1},$$

where $u_a \in R^\times$. Symbols all of whose entries are units have zero residue. If $\pi' = \epsilon \pi$ is another uniformizer, then

$$\{\pi', u_1, \dots, u_{i-1}\} = \{\pi, u_1, \dots, u_{i-1}\} + \{\epsilon, u_1, \dots, u_{i-1}\},$$

and the second summand has zero residue; hence the displayed description is independent of the chosen uniformizer. For $i = 1$, the same formula reads $\partial_R\{\pi\} = 1 \in W_n\Omega_{\bar{k},\log}^0$. For $i \geq 2$ and $\bar{k}$ separably closed the target $W_n\Omega_{\bar{k},\log}^{i-1}$ is zero, but the same formula is the sheaf-level normalization before taking geometric stalks.

The long exact sequence therefore gives

$$H_{\bar{s}}^q(\mathrm{Spec} R, T) = 0 \quad (q \leq i), \quad H_{\bar{s}}^{i+1}(\mathrm{Spec} R, T) \simeq W_n\Omega_{\bar{k},\log}^{i-1}.$$

For $q > i + 1$, the same sequence identifies

$$H_{\bar{s}}^q(\mathrm{Spec} R, T) \simeq H^{q-1}(L, \mu_{p^n}^{\otimes i}).$$

The field L is the fraction field of a henselian discrete valuation ring with separably closed residue field, hence has p -cohomological dimension 1. Thus the last group vanishes for $q > i + 1$, since then $q - 1 \geq 2$. Serre's cohomological dimension theorem for henselian discretely valued fields gives $\mathrm{cd}_p(L) = \mathrm{cd}_p(\bar{k}) + 1 = 1$; see [8, Chapter II, §4].

Consequently $R\iota^!\mathfrak{I}_n(i)_X$ has a single nonzero cohomology sheaf, namely $W_n\Omega_{s,\log}^{i-1}$ in degree $i + 1$. By Definition 1.1,

$$\mathfrak{I}_n(i-1)_s[-2] = W_n\Omega_{s,\log}^{i-1}[-(i-1)-2] = W_n\Omega_{s,\log}^{i-1}[-i-1],$$

which is exactly the complex just computed. This proves

$$\iota^!\mathfrak{I}_n(i)_X \simeq \mathfrak{I}_n(i-1)_s[-2].$$

The displayed residue formula gives the stated normalization of the isomorphism. $\square$

Proposition 1.6 (Comparison and localization on $\mathrm{Spec} \mathcal{O}_K$). *Let $X = \mathrm{Spec} \mathcal{O}_K$, let $s = \mathrm{Spec} k$, let*

$$j : \mathrm{Spec} K \hookrightarrow X, \quad \iota : s \hookrightarrow X$$

be the generic and closed immersions. For every $i \geq 0$, there is a canonical equivalence

$$\mathbb{Z}_p(i)_X^{\mathrm{syn}} \simeq \mathbb{Z}_p(i)_X^{\mathrm{GSS}}$$

on $X_{\mathrm{\acute{e}t}}$. Hence

$$\mathbb{Z}_p(i)(\mathcal{O}_K) \simeq \mathrm{R}\Gamma(X_{\mathrm{\acute{e}t}}, \mathbb{Z}_p(i)_X^{\mathrm{GSS}}).$$

For every $i \geq 1$, there is a localization triangle

$$\mathrm{R}\Gamma(k, \mathbb{Z}_p(i-1))[-2] \longrightarrow \mathbb{Z}_p(i)(\mathcal{O}_K) \longrightarrow \mathrm{R}\Gamma(K, \mathbb{Z}_p(i)) \xrightarrow{+1},$$

where the first term is computed from $\mathbb{Z}_p(i-1)_s^{\mathrm{GSS}}$, and in the third term $\mathbb{Z}_p(i)$ is the rank-one continuous $\mathbb{Z}_p$ -module on which $g \in G_K$ acts by $\chi_K(g)^i$.

Proof. We use Bhatt–Mathew in its finite-level Sato-comparison form. Their Theorem 1.8 gives the nearby-cycle description of $\mathbb{Z}/p^n\mathbb{Z}(i)_X^{\mathrm{syn}}$: it agrees with $\tau_{\leq i}Rj_*\mu_{p^n}^{\otimes i}$ below degree i , and in degree i its image is the subsheaf generated by symbols. Their Theorem 5.8 then identifies this finite-level syntomic complex with Sato's $\mathfrak{I}_n(i)_X$ for regular semistable schemes over a Dedekind domain whose characteristic p residue fields are perfect.

In the present case take the base Dedekind domain to be $A = \mathcal{O}_K$. Then $X = \mathrm{Spec} \mathcal{O}_K$ is regular and finite type over A , the only characteristic p residue field of A is the finite residue field k , hence perfect, and $X \rightarrow \mathrm{Spec} \mathcal{O}_K$ is the identity regular trait, hence semistable. Thus the hypotheses of Theorem 5.8 apply. The comparison is canonical and multiplicative, and is compatible with the reduction maps $\mathbb{Z}/p^{n+1}\mathbb{Z} \rightarrow \mathbb{Z}/p^n\mathbb{Z}$: on the generic fiber these maps are the reduction maps of Kummer twists, and on the top cohomology sheaf they are the reductions

of the same symbol subsheaf and residue construction. Hence the finite-level comparisons form an equivalence of inverse systems. Applying $R\varprojlim_n$ gives

$$\mathbb{Z}_p(i)_X^{\text{syn}} \simeq R\varprojlim_n \mathfrak{I}_n(i)_X = \mathbb{Z}_p(i)_X^{\text{GSS}}.$$

Taking étale cohomology on X gives the displayed identification of global sections. See [1, Theorems 1.8 and 5.8].

We next spell out the localization triangle. Work first with $\mathbb{Z}/p^n\mathbb{Z}$ -coefficients and put $F_n = \mathfrak{I}_n(i)_X$. The open-closed decomposition

$$s \xrightarrow{\iota} X \xleftarrow{j} \text{Spec } K$$

gives the localization triangle

$$R\Gamma_s(X, F_n) \longrightarrow R\Gamma(X, F_n) \longrightarrow R\Gamma(K, j^*F_n) \xrightarrow{+1}.$$

The term with support on s is computed by extraordinary pullback:

$$R\Gamma_s(X, F_n) \simeq R\Gamma(s, \iota^!F_n).$$

By Lemma 1.5,

$$\iota^!\mathfrak{I}_n(i)_X \simeq \mathfrak{I}_n(i-1)_s[-2], \quad j^*\mathfrak{I}_n(i)_X \simeq \mu_{p^n}^{\otimes i}.$$

Thus the finite-level triangle is

$$R\Gamma(s, \mathfrak{I}_n(i-1)_s)[-2] \longrightarrow R\Gamma(X, \mathfrak{I}_n(i)_X) \longrightarrow R\Gamma(K, \mu_{p^n}^{\otimes i}) \xrightarrow{+1}.$$

These finite-level triangles and identifications are functorial in n , so they form a distinguished triangle in the derived category of inverse systems. Applying the exact functor $R\varprojlim_n$ to that triangle gives

$$R\Gamma(s, \mathbb{Z}_p(i-1)_s^{\text{GSS}})[-2] \longrightarrow R\Gamma(X, \mathbb{Z}_p(i)_X^{\text{GSS}}) \longrightarrow R\Gamma(K, \mathbb{Z}_p(i)) \xrightarrow{+1};$$

here

$$R\Gamma(K, \mathbb{Z}_p(i)) := R\varprojlim_n R\Gamma(K, \mu_{p^n}^{\otimes i})$$

is continuous p -adic Galois cohomology of the rank-one module described in the statement, in the sense of Jannsen. No ordinary inverse-limit exactness is being assumed here; possible derived-limit terms are part of this definition. Replacing the middle term by $\mathbb{Z}_p(i)(\mathcal{O}_K)$ using the first part gives the asserted triangle.

It remains only to record the closed-fiber object used later. By Definition 1.1,

$$\mathfrak{I}_n(m)_s \simeq W_n\Omega_{s,\log}^m[-m]$$

for $m \geq 0$, where $W_n\Omega_{s,\log}^0 = \mathbb{Z}/p^n\mathbb{Z}$. Thus

$$\mathbb{Z}_p(m)_s^{\text{GSS}} = R\varprojlim_n W_n\Omega_{s,\log}^m[-m].$$

This is the special-fiber object used in the finite-field calculation below. $\square$

The proof of Theorem 1.2 uses the comparison proposition above and the cohomological inputs collected in Section 2. The proof is given in Section 3. After the proof, Sections 4 and 5 compute the finite groups $W_m(K)$ that occur in the statement, and Section 6 records the derived splitting and explicit consequences.

2 Cohomological inputs

This section isolates the cohomological facts used in the proof of Theorem 1.2: the finite-field term in the localization triangle, the two residue maps in weight one, and the local Galois cohomology of positive Tate twists.

Lemma 2.1 (Finite fields). *Let k be a finite field of characteristic p . Then*

$$H^0(k, \mathbb{Z}_p) = \mathbb{Z}_p, \quad H^1(k, \mathbb{Z}_p) = \mathbb{Z}_p, \quad H^r(k, \mathbb{Z}_p) = 0 \quad (r \geq 2).$$

Moreover, for $m \geq 1$,

$$R\Gamma(k, \mathbb{Z}_p(m)_k^{\text{GSS}}) \simeq 0.$$

In the localization triangles below this last complex is abbreviated to $R\Gamma(k, \mathbb{Z}_p(m))$.

Proof. The first assertions concern the constant sheaf $\mathbb{Z}_p$. Write $G_k = \text{Gal}(\bar{k}/k)$. It is topologically generated by Frobenius and is isomorphic to $\widehat{\mathbb{Z}}$. For finite coefficients $\mathbb{Z}/p^r\mathbb{Z}$ with trivial action,

$$H^0(k, \mathbb{Z}/p^r\mathbb{Z}) = \mathbb{Z}/p^r\mathbb{Z}, \quad H^1(k, \mathbb{Z}/p^r\mathbb{Z}) = \text{Hom}_{\text{cts}}(G_k, \mathbb{Z}/p^r\mathbb{Z}) = \mathbb{Z}/p^r\mathbb{Z},$$

and $H^q(k, \mathbb{Z}/p^r\mathbb{Z}) = 0$ for $q \geq 2$, because the p -cohomological dimension of G_k is 1. Passing to the inverse limit over r gives

$$H^0(k, \mathbb{Z}_p) = \mathbb{Z}_p, \quad H^1(k, \mathbb{Z}_p) = \mathbb{Z}_p, \quad H^q(k, \mathbb{Z}_p) = 0 \quad (q \geq 2).$$

For $m \geq 1$, the special-fiber identification in Proposition 1.6 gives

$$\mathbb{Z}_p(m)_k^{\text{GSS}} \simeq R\varprojlim_n W_n \Omega_{k, \log}^m[-m].$$

Write $q = |k|$. By the logarithmic Hodge–Witt convention recalled in Definition 1.1, equivalently by the Bloch–Kato symbol description cited there, the sheaf $W_n \Omega_{k, \log}^m$ is generated by symbols

$$d \log[a_1] \wedge \cdots \wedge d \log[a_m], \quad a_j \in k^\times.$$

For every $a \in k^\times$, $a^{q-1} = 1$, and $q-1$ is prime to p . Therefore

$$0 = d \log[a^{q-1}] = (q-1) d \log[a].$$

Multiplication by $q-1$ is invertible on the p^n -torsion sheaf $W_n \Omega_{k, \log}^1$, so $d \log[a] = 0$ for every a . Hence the sheaf $W_n \Omega_{k, \log}^m$ itself is zero for $m \geq 1$ and every n . Therefore each finite-level complex $R\Gamma(k, W_n \Omega_{k, \log}^m[-m])$ is zero, and so $R\Gamma(k, \mathbb{Z}_p(m)) \simeq 0$ for $m \geq 1$. $\square$

Lemma 2.2 (Weight-one residue maps). *In the localization triangle*

$$R\Gamma(k, \mathbb{Z}_p)[-2] \longrightarrow C_1 \longrightarrow R\Gamma(K, \mathbb{Z}_p(1)) \xrightarrow{+1},$$

the boundary map

$$H^1(K, \mathbb{Z}_p(1)) \longrightarrow H^0(k, \mathbb{Z}_p)$$

is the p -completed valuation on $K^\times$. The boundary map

$$H^2(K, \mathbb{Z}_p(1)) \longrightarrow H^1(k, \mathbb{Z}_p)$$

is the inverse limit of the Brauer residue maps and is an isomorphism.

Proof. Work first with $\mathbb{Z}/p^n\mathbb{Z}$ -coefficients. Lemma 1.5 gives the finite-level localization triangle

$$R\Gamma(k, \mathbb{Z}/p^n\mathbb{Z})[-2] \longrightarrow R\Gamma(X, \mathfrak{I}_n(1)_X) \longrightarrow R\Gamma(K, \mu_{p^n}) \xrightarrow{+1}.$$

The boundary map

$$H^1(K, \mu_{p^n}) \longrightarrow H^0(k, \mathbb{Z}/p^n\mathbb{Z})$$

is the degree-one part of the residue in Lemma 1.5. Under the Kummer isomorphism

$$K^\times / (K^\times)^{p^n} \cong H^1(K, \mu_{p^n}),$$

write $x = \pi^{v_K(x)}u$ with $u \in \mathcal{O}_K^\times$. The residue normalization gives $\{\pi\} \mapsto 1$ and unit symbols map to 0. Hence this boundary sends x to $v_K(x) \bmod p^n$. Passing to $R\varprojlim_n$ gives the p -completed valuation

$$\widehat{K^\times}_p \longrightarrow \mathbb{Z}_p.$$

In degree two, the same finite-level localization sequence gives the p^n -primary Brauer residue. Under the Kummer sequence,

$$H^2(K, \mu_{p^n}) = \mathrm{Br}(K)[p^n] \longrightarrow H^1(k, \mathbb{Z}/p^n\mathbb{Z}).$$

This is the Brauer residue for a henselian discretely valued field: Sato's product and residue compatibility identify the localization boundary with the classical Brauer residue, and the normalization is the same as above on cyclic algebras generated by a uniformizer. For a henselian discretely valued field with residue field k , the Brauer residue sequence is

$$0 \longrightarrow \mathrm{Br}(\mathcal{O}_K)[p^n] \longrightarrow \mathrm{Br}(K)[p^n] \longrightarrow H^1(k, \mathbb{Z}/p^n\mathbb{Z}) \longrightarrow 0.$$

Moreover $\mathrm{Br}(\mathcal{O}_K) \cong \mathrm{Br}(k) = 0$, because k is finite. Hence this residue map is an isomorphism for every n . These residue maps are compatible with the transition maps in n ; taking $R\varprojlim_n$ gives an isomorphism

$$H^2(K, \mathbb{Z}_p(1)) \xrightarrow{\sim} H^1(k, \mathbb{Z}_p).$$

The compatibility with Sato's localization, purity, and products is part of the finite-level construction of $\mathfrak{I}_n(1)$; see [6, Theorem 1.1.1, especially T3 and T4, and §4.2]. For the classical Brauer residue sequence, see [7, Chapter XIII]. $\square$

Lemma 2.3 (Positive Tate twists). *Let $i \geq 2$. Then*

$$H^0(K, \mathbb{Z}_p(i)) = 0, \quad H^2(K, \mathbb{Z}_p(i)) \cong W_{1-i}(K)^\vee.$$

Furthermore $H^1(K, \mathbb{Z}_p(i))$ is a finitely generated $\mathbb{Z}_p$ -module of rank n_K , with torsion subgroup $W_i(K)$. Consequently, noncanonically,

$$H^1(K, \mathbb{Z}_p(i)) \cong \mathbb{Z}_p^{n_K} \oplus W_i(K).$$

Proof. The image $H_K = \chi_K(G_K)$ is open in $\mathbb{Z}_p^\times$. Since $i \neq 0$, the character χ_K^i is nontrivial on this open image. If $x \in \mathbb{Z}_p(i)$ were fixed by G_K , then $h^i x = x$ for all $h \in H_K$. Choosing h with $h^i \neq 1$ forces $x = 0$, because $\mathbb{Z}_p$ is torsion-free. Hence $H^0(K, \mathbb{Z}_p(i)) = 0$.

Tate local duality gives

$$H^2(K, \mathbb{Z}_p(i)) \cong H^0(K, \mathbb{Q}_p/\mathbb{Z}_p(1-i))^\vee = W_{1-i}(K)^\vee.$$

To compute H^1 , use the exact sequence

$$0 \longrightarrow \mathbb{Z}_p(i) \longrightarrow \mathbb{Q}_p(i) \longrightarrow \mathbb{Q}_p/\mathbb{Z}_p(i) \longrightarrow 0.$$

Since $H^0(K, \mathbb{Q}_p(i)) = 0$, the connecting map in cohomology injects

$$W_i(K) = H^0(K, \mathbb{Q}_p/\mathbb{Z}_p(i)) \hookrightarrow H^1(K, \mathbb{Z}_p(i)).$$

The quotient of $H^1(K, \mathbb{Z}_p(i))$ by this image injects into the $\mathbb{Q}_p$ -vector space $H^1(K, \mathbb{Q}_p(i))$. A subgroup of a $\mathbb{Q}_p$ -vector space has no nonzero $\mathbb{Z}_p$ -torsion, so the displayed finite subgroup is precisely the torsion subgroup of $H^1(K, \mathbb{Z}_p(i))$.

For $V = \mathbb{Q}_p(i)$, $i \geq 2$, one has

$$H^0(K, V) = 0, \quad H^2(K, V) = H^0(K, \mathbb{Q}_p(1-i))^* = 0.$$

The local Euler characteristic formula therefore gives

$$\dim_{\mathbb{Q}_p} H^1(K, V) = n_K.$$

The finiteness theorem for local Galois cohomology, together with comparison after tensoring with $\mathbb{Q}_p$, gives

$$H^1(K, \mathbb{Z}_p(i)) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p \cong H^1(K, V).$$

These are the finiteness and scalar-extension statements for local Galois cohomology of p -adic representations; see [5, Chapter I]. Thus $H^1(K, \mathbb{Z}_p(i))$ is finitely generated over $\mathbb{Z}_p$, and its torsion-free quotient has rank n_K . A finitely generated torsion-free module over the discrete valuation ring $\mathbb{Z}_p$ is free, and the short exact sequence

$$0 \longrightarrow W_i(K) \longrightarrow H^1(K, \mathbb{Z}_p(i)) \longrightarrow \mathbb{Z}_p^{n_K} \longrightarrow 0$$

splits noncanonically because $\mathbb{Z}_p^{n_K}$ is projective. □

3 Proof of the syntomic theorem

This section proves Theorem 1.2. For $i \geq 1$, Proposition 1.6 gives the localization triangle

$$\mathrm{R}\Gamma(k, \mathbb{Z}_p(i-1))[-2] \longrightarrow C_i \longrightarrow \mathrm{R}\Gamma(K, \mathbb{Z}_p(i)) \xrightarrow{+1}.$$

3.1 Negative weights

For $i < 0$, the BMS normalization gives $\mathbb{Z}_p(i) = 0$; see [3, Theorem 1.12 and Remark 1.14]. Hence $C_i \simeq 0$.

3.2 Weight zero

The same normalization identifies $\mathbb{Z}_p(0)$ with the constant sheaf $\mathbb{Z}_p$. Since $\mathcal{O}_K$ is henselian with residue field k , finite étale covers of $\mathrm{Spec} \mathcal{O}_K$ are equivalent to finite étale covers of $\mathrm{Spec} k$. Thus

$$\mathrm{R}\Gamma(\mathcal{O}_K, \mathbb{Z}_p) \simeq \mathrm{R}\Gamma(k, \mathbb{Z}_p).$$

Lemma 2.1 gives

$$H^0(C_0) = \mathbb{Z}_p, \quad H^1(C_0) = \mathbb{Z}_p,$$

and no other cohomology.

3.3 Weight one

For $i = 1$, the preceding localization triangle is

$$\mathrm{R}\Gamma(k, \mathbb{Z}_p)[-2] \longrightarrow C_1 \longrightarrow \mathrm{R}\Gamma(K, \mathbb{Z}_p(1)) \xrightarrow{+1}.$$

The corresponding part of the long exact sequence is

$$\begin{aligned} 0 \rightarrow H^1(C_1) \rightarrow H^1(K, \mathbb{Z}_p(1)) &\xrightarrow{\partial_1} H^0(k, \mathbb{Z}_p) \rightarrow H^2(C_1) \\ \rightarrow H^2(K, \mathbb{Z}_p(1)) &\xrightarrow{\partial_2} H^1(k, \mathbb{Z}_p) \rightarrow H^3(C_1) \rightarrow 0. \end{aligned}$$

The zeroes at the two ends come from $H^0(K, \mathbb{Z}_p(1)) = 0$, the fact that $\mathrm{R}\Gamma(k, \mathbb{Z}_p)[-2]$ has no cohomology below degree 2, $H^2(k, \mathbb{Z}_p) = 0$, and the vanishing $H^3(K, \mathbb{Z}_p(1)) = 0$ from the p -cohomological dimension 2 of G_K . Kummer theory identifies

$$H^1(K, \mathbb{Z}_p(1)) \cong \widehat{K^\times}_p.$$

By Lemma 2.2, under this identification ∂_1 is the p -completed valuation

$$\widehat{K^\times}_p \longrightarrow \mathbb{Z}_p.$$

Since $K^\times \cong \varpi^\mathbb{Z} \times \mathcal{O}_K^\times$, this map is surjective and its kernel is $\widehat{\mathcal{O}_{Kp}^\times}$. Hence

$$H^1(C_1) \cong \widehat{\mathcal{O}_{Kp}^\times}.$$

Lemma 2.2 also shows that ∂_2 is an isomorphism. Since ∂_1 is surjective and ∂_2 is an isomorphism, $H^2(C_1) = H^3(C_1) = 0$; the preceding terms give $H^0(C_1) = 0$.

Finally, use the exact sequence

$$1 \longrightarrow U_1 \longrightarrow \mathcal{O}_K^\times \longrightarrow k^\times \longrightarrow 1, \quad U_1 = 1 + \mathfrak{m}_K.$$

The group $k^\times$ has order prime to p , so its p -completion is zero and

$$\widehat{\mathcal{O}_{Kp}^\times} \cong \widehat{U_{1p}}.$$

The torsion subgroup of U_1 is $\mu_{p^\infty}(K)$. Choose N so large that $1 + \mathfrak{m}_K^N$ contains no nontrivial element of this finite torsion subgroup and the p -adic logarithm is defined on it. Then

$$\log : 1 + \mathfrak{m}_K^N \xrightarrow{\sim} \mathfrak{m}_K^N$$

identifies $1 + \mathfrak{m}_K^N$ with a free $\mathbb{Z}_p$ -module of rank n_K , and $U_1/(1 + \mathfrak{m}_K^N)$ is finite. Therefore $\widehat{U_{1p}}$ is a finitely generated $\mathbb{Z}_p$ -module whose torsion subgroup is $\mu_{p^\infty}(K)$ and whose torsion-free quotient has rank n_K . Since $\mathbb{Z}_p$ is a principal ideal domain, the quotient is free and the short exact sequence splits after choosing a complement. Thus

$$\widehat{\mathcal{O}_{Kp}^\times} \cong \mathbb{Z}_p^{n_K} \oplus \mu_{p^\infty}(K)$$

noncanonically.

3.4 Weights at least two

Let $i \geq 2$. The preceding localization triangle gives

$$\mathrm{R}\Gamma(k, \mathbb{Z}_p(i-1))[-2] \longrightarrow C_i \longrightarrow \mathrm{R}\Gamma(K, \mathbb{Z}_p(i)) \xrightarrow{+1}.$$

Since $i-1 \geq 1$, Lemma 2.1 gives $\mathrm{R}\Gamma(k, \mathbb{Z}_p(i-1)) \simeq 0$. Therefore

$$C_i \simeq \mathrm{R}\Gamma(K, \mathbb{Z}_p(i)).$$

The computation of the cohomology groups is Lemma 2.3. This completes the proof of Theorem 1.2.

The proof just completed leaves the finite groups $W_m(K)$ in cohomological form. We now compute these groups from the image of the cyclotomic character and then substitute the resulting formulae into the syntomic calculation.

4 Cyclotomic torsion groups

Let

$$\chi_K : G_K \longrightarrow \mathbb{Z}_p^\times$$

be the p -adic cyclotomic character, normalized so that

$$g(\zeta) = \zeta^{\chi_K(g) \bmod p^n}$$

for $\zeta \in \mu_{p^n}(\overline{K})$. Put

$$H_K := \chi_K(G_K) \subset \mathbb{Z}_p^\times.$$

The subgroup H_K is open in $\mathbb{Z}_p^\times$. Equivalently, the fixed field of H_K inside $\mathbb{Q}_p(\mu_{p^\infty})$ is $K \cap \mathbb{Q}_p(\mu_{p^\infty})$, which is a finite extension of $\mathbb{Q}_p$. This is the cyclotomic case of local class field theory; see [7, Chapter XV].

Definition 4.1 (Cyclotomic depth). *For $i \neq 0$, define*

$$a_i(K) := \max\{n \geq 0 \mid h^i \equiv 1 \pmod{p^n} \text{ for every } h \in H_K\}.$$

Equivalently,

$$a_i(K) = \inf_{h \in H_K} v_p(h^i - 1),$$

where $v_p(i)$ for a nonzero integer i means $v_p(|i|)$, and with the convention that the value is 0 if some $h^i - 1$ is a p -adic unit.

Lemma 4.2. *For $i \neq 0$, the integer $a_i(K)$ is finite.*

Proof. Since H_K is open in $\mathbb{Z}_p^\times$, its image in the torsion-free pro- p factor is an open subgroup. If p is odd, this factor is $P = 1 + p\mathbb{Z}_p$; if $p = 2$, it is $P = 1 + 4\mathbb{Z}_2$. Let

$$\pi : \mathbb{Z}_p^\times \longrightarrow P$$

be the projection in the decomposition $\mathbb{Z}_p^\times = \mu_{p-1} \times P$, for odd p , or $\mathbb{Z}_2^\times = \{\pm 1\} \times P$, for $p = 2$. Choose $h \in H_K$ such that $u = \pi(h) \neq 1$. The p -adic logarithm is injective on P , and

$$\log(u^i) = i \log(u).$$

Since $i \neq 0$ and $\log(u) \neq 0$, one has $u^i \neq 1$. The pro- p projection of h^i is u^i , hence $h^i \neq 1$. Thus $v_p(h^i - 1) < \infty$, so the infimum in Definition 4.1 is finite. $\square$

Theorem 4.3 (Intrinsic computation of $W_i(K)$). *For $i = 0$,*

$$W_0(K) \cong \mathbb{Q}_p/\mathbb{Z}_p.$$

For $i \neq 0$, the fixed subgroup of the cyclic p -divisible group $\mathbb{Q}_p/\mathbb{Z}_p(i)$ is

$$W_i(K) = (\mathbb{Q}_p/\mathbb{Z}_p(i))[p^{a_i(K)}].$$

In particular, as an abstract abelian group,

$$W_i(K) \cong \mathbb{Z}/p^{a_i(K)}\mathbb{Z},$$

where $\mathbb{Z}/p^0\mathbb{Z}$ means the zero group.

Proof. The case $i = 0$ is immediate. Assume $i \neq 0$. For $n \geq 1$, the p^n -torsion subgroup of $\mathbb{Q}_p/\mathbb{Z}_p(i)$ is

$$(\mathbb{Q}_p/\mathbb{Z}_p(i))[p^n] \cong \mathbb{Z}/p^n\mathbb{Z}(i),$$

and $g \in G_K$ acts on it by multiplication by $\chi_K(g)^i$. Thus the full subgroup $(\mathbb{Q}_p/\mathbb{Z}_p(i))[p^n]$ is fixed pointwise by G_K if and only if

$$h^i \equiv 1 \pmod{p^n} \quad \text{for every } h \in H_K.$$

This holds precisely for $n \leq a_i(K)$. Since the subgroups of $\mathbb{Q}_p/\mathbb{Z}_p(i)$ form the chain

$$0 \subset (\mathbb{Q}_p/\mathbb{Z}_p(i))[p] \subset (\mathbb{Q}_p/\mathbb{Z}_p(i))[p^2] \subset \cdots \subset \mathbb{Q}_p/\mathbb{Z}_p(i),$$

the fixed subgroup is the largest full p^n -torsion subgroup on which the action is trivial, namely $(\mathbb{Q}_p/\mathbb{Z}_p(i))[p^{a_i(K)}]$. $\square$

Corollary 4.4 (Finite-level form). *Let*

$$G_{K,n} := \text{im}(G_K \xrightarrow{\chi_K} (\mathbb{Z}/p^n\mathbb{Z})^\times) \cong \text{Gal}(K(\mu_{p^n})/K).$$

For $n = 0$, interpret the congruence modulo p^0 as vacuous and $G_{K,0}$ as the trivial group. For $i \neq 0$,

$$a_i(K) = \max\{n \geq 0 \mid x^i = 1 \text{ for all } x \in G_{K,n}\}.$$

Equivalently, $p^{a_i(K)}$ is the largest order of a finite subgroup

$$\mathbb{Z}/p^n\mathbb{Z}(i) \subset \mathbb{Q}_p/\mathbb{Z}_p(i)$$

on which G_K acts trivially.

Proof. The action of G_K on $\mathbb{Z}/p^n\mathbb{Z}(i)$ factors through the image $G_{K,n}$ of the cyclotomic character modulo p^n . The full subgroup $\mathbb{Z}/p^n\mathbb{Z}(i)$ is fixed pointwise if and only if every element $x \in G_{K,n}$ acts by multiplication by $x^i = 1$. Taking the largest such n gives exactly the same maximum as in Definition 4.1. The last assertion is then Theorem 4.3 restricted to finite level. $\square$

5 Closed formulae for $W_i(K)$

5.1 Odd primes

Assume in this subsection that p is odd. Use the canonical decomposition

$$\mathbb{Z}_p^\times = \mu_{p-1} \times (1 + p\mathbb{Z}_p),$$

and write

$$\omega : \mathbb{Z}_p^\times \rightarrow \mu_{p-1}, \quad \langle \cdot \rangle : \mathbb{Z}_p^\times \rightarrow 1 + p\mathbb{Z}_p$$

for the two projections.

Definition 5.1. *Let*

$$d_K := |\omega(H_K)|.$$

Equivalently, $d_K = [K(\mu_p) : K]$, because the mod p cyclotomic image is $\text{Gal}(K(\mu_p)/K)$. Since $\langle H_K \rangle$ is an open subgroup of $1 + p\mathbb{Z}_p$, there is a unique integer $c_K \geq 1$ such that

$$\langle H_K \rangle = 1 + p^{c_K}\mathbb{Z}_p.$$

Lemma 5.2. *Let p be odd, let $c \geq 1$, and let $i \neq 0$. Then*

$$\inf_{u \in 1 + p^c\mathbb{Z}_p} v_p(u^i - 1) = c + v_p(i).$$

Proof. The p -adic logarithm gives an isomorphism

$$\log : 1 + p^c\mathbb{Z}_p \xrightarrow{\sim} p^c\mathbb{Z}_p.$$

For $u \in 1 + p^c\mathbb{Z}_p$, one has $\log(u^i) = i \log(u)$, also for negative i , and $v_p(u^i - 1) = v_p(\log(u^i))$ because $u^i \in 1 + p\mathbb{Z}_p$. Therefore

$$v_p(u^i - 1) = v_p(i) + v_p(\log u).$$

As u ranges over $1 + p^c\mathbb{Z}_p$, the valuation of $\log u$ ranges over all integers at least c , and the minimum c is attained. $\square$

Theorem 5.3 (Odd prime formula). *Let p be odd and $i \neq 0$. Then*

$$a_i(K) = \begin{cases} 0, & d_K \nmid i, \\ c_K + v_p(i), & d_K \mid i, \end{cases}$$

and therefore

$$W_i(K) \cong \begin{cases} 0, & d_K \nmid i, \\ \mathbb{Z}/p^{c_K + v_p(i)}\mathbb{Z}, & d_K \mid i. \end{cases}$$

Proof. Write $h = \omega(h)\langle h \rangle$ for $h \in H_K$. If $d_K \nmid i$, then $\omega(H_K)$ contains an element ζ with $\zeta^i \neq 1$. Lifting it to $h \in H_K$ gives $h^i \not\equiv 1 \pmod{p}$, hence $a_i(K) = 0$.

If $d_K \mid i$, then $\omega(h)^i = 1$ for every $h \in H_K$. Hence $h^i = \langle h \rangle^i$, and the possible valuations of $h^i - 1$ are exactly the possible valuations of $u^i - 1$ for $u \in \langle H_K \rangle = 1 + p^{c_K}\mathbb{Z}_p$. Lemma 5.2 gives $a_i(K) = c_K + v_p(i)$. The description of $W_i(K)$ follows from Theorem 4.3. $\square$

Example 5.4 (The field $\mathbb{Q}_p$, p odd). *For $K = \mathbb{Q}_p$, $H_K = \mathbb{Z}_p^\times$, so $d_K = p - 1$ and $c_K = 1$. Hence, for $i \neq 0$,*

$$W_i(\mathbb{Q}_p) \cong \begin{cases} 0, & p - 1 \nmid i, \\ \mathbb{Z}/p^{1 + v_p(i)}\mathbb{Z}, & p - 1 \mid i. \end{cases}$$

In particular, $W_{p-1}(\mathbb{Q}_p) \cong \mathbb{Z}/p\mathbb{Z}$.

Example 5.5 (The first twist). *For every K and every prime p ,*

$$W_1(K) = H^0(K, \mathbb{Q}_p/\mathbb{Z}_p(1)) = \mu_{p^\infty}(K).$$

For odd p , Theorem 5.3 says that this group is nonzero exactly when K contains μ_p .

5.2 The prime 2

Assume now that $p = 2$. Use the canonical decomposition

$$\mathbb{Z}_2^\times = \{\pm 1\} \times (1 + 4\mathbb{Z}_2),$$

and write

$$\varepsilon : \mathbb{Z}_2^\times \rightarrow \{\pm 1\}, \quad \gamma : \mathbb{Z}_2^\times \rightarrow 1 + 4\mathbb{Z}_2$$

for the two projections.

Definition 5.6. *Let*

$$s_K := \begin{cases} 0, & \varepsilon(H_K) = \{1\}, \\ 1, & \varepsilon(H_K) = \{\pm 1\}. \end{cases}$$

Since $\gamma(H_K)$ is an open subgroup of $1 + 4\mathbb{Z}_2$, there is a unique integer $b_K \geq 2$ such that

$$\gamma(H_K) = 1 + 2^{b_K}\mathbb{Z}_2.$$

Lemma 5.7. *Let $b \geq 2$ and $i \neq 0$. Then*

$$\inf_{u \in 1 + 2^b\mathbb{Z}_2} v_2(u^i - 1) = b + v_2(i).$$

Proof. The 2-adic logarithm gives an isomorphism

$$\log : 1 + 4\mathbb{Z}_2 \xrightarrow{\sim} 4\mathbb{Z}_2,$$

and restricts to an isomorphism $1 + 2^b\mathbb{Z}_2 \simeq 2^b\mathbb{Z}_2$ for $b \geq 2$. For $u \in 1 + 2^b\mathbb{Z}_2$, the element u^i still lies in $1 + 4\mathbb{Z}_2$, and $\log(u^i) = i \log(u)$. As in Lemma 5.2,

$$v_2(u^i - 1) = v_2(i) + v_2(\log u),$$

and the minimum of $v_2(\log u)$ on $1 + 2^b\mathbb{Z}_2$ is b . □

Theorem 5.8 (The 2-adic formula). *Let $p = 2$ and $i \neq 0$. Then*

$$W_i(K) \cong \mathbb{Z}/2^{a_i(K)}\mathbb{Z},$$

where

$$a_i(K) = \begin{cases} 1, & s_K = 1 \text{ and } i \text{ is odd,} \\ b_K + v_2(i), & \text{otherwise.} \end{cases}$$

Proof. Write $h = \varepsilon(h)\gamma(h)$ for $h \in H_K$. Suppose first that $s_K = 1$ and i is odd. Choose $h \in H_K$ with $\varepsilon(h) = -1$. Since $\gamma(h)^i \in 1 + 4\mathbb{Z}_2$,

$$h^i - 1 = -\gamma(h)^i - 1 = -(\gamma(h)^i + 1),$$

and $\gamma(h)^i + 1 \equiv 2 \pmod{4}$. Thus $v_2(h^i - 1) = 1$. Conversely, every odd 2-adic unit is congruent to 1 modulo 2, so $v_2(h^i - 1) \geq 1$ for all $h \in H_K$. Therefore $a_i(K) = 1$.

In all other cases, either $s_K = 0$, so $\varepsilon(h) = 1$ for every $h \in H_K$, or i is even, so $\varepsilon(h)^i = 1$ for every $h \in H_K$. The congruence depth is then controlled by $\gamma(h)^i - 1$. Since $\gamma(H_K) = 1 + 2^{b_K}\mathbb{Z}_2$, Lemma 5.7 gives $a_i(K) = b_K + v_2(i)$. The result follows from Theorem 4.3. For $i = 1$, this says in particular that $W_1(K) = \mu_{2^\infty}(K)$ has order 2 when the sign image is nontrivial, while if $s_K = 0$ (equivalently K contains μ_4) its order is 2^{b_K} . □

Example 5.9 (The field $\mathbb{Q}_2$). *For $K = \mathbb{Q}_2$, $H_K = \mathbb{Z}_2^\times$, so $s_K = 1$ and $b_K = 2$. Hence*

$$W_i(\mathbb{Q}_2) \cong \begin{cases} \mathbb{Z}/2\mathbb{Z}, & i \text{ odd,} \\ \mathbb{Z}/2^{2+v_2(i)}\mathbb{Z}, & i \text{ even and } i \neq 0. \end{cases}$$

In particular $W_1(\mathbb{Q}_2) = \mu_{2^\infty}(\mathbb{Q}_2) = \{\pm 1\} \cong \mathbb{Z}/2\mathbb{Z}$.

6 Derived splitting and explicit consequences

This final section turns the cohomology calculation into derived-category decompositions and records the explicit forms obtained by substituting the cyclotomic torsion formulae.

Lemma 6.1. *Let R be a principal ideal domain. Every bounded complex of R -modules is noncanonically quasi-isomorphic to the direct sum of its cohomology groups placed in their cohomological degrees.*

Proof. A principal ideal domain has global dimension 1, so $\text{Ext}_R^q(-, -) = 0$ for $q \geq 2$. For a bounded complex C , the truncation triangles

$$\tau_{\leq n-1}C \longrightarrow \tau_{\leq n}C \longrightarrow H^n(C)[-n] \xrightarrow{+1}$$

are classified by obstruction classes lying in second Ext groups after the lower truncations have been split. These obstruction groups vanish, and induction on the length of C gives the claim. $\square$

Proof of Corollary 1.3. Apply Lemma 6.1 with $R = \mathbb{Z}_p$. The cohomology groups of each C_i are those listed in Theorem 1.2. Placing each nonzero cohomology group in its cohomological degree gives

$$\begin{aligned} C_0 &\simeq H^0(C_0) \oplus H^1(C_0)[-1] \simeq \mathbb{Z}_p \oplus \mathbb{Z}_p[-1], \\ C_1 &\simeq H^1(C_1)[-1] \simeq \widehat{\mathcal{O}_{Kp}^\times}[-1], \end{aligned}$$

and, for $i \geq 2$,

$$C_i \simeq H^1(K, \mathbb{Z}_p(i))[-1] \oplus W_{1-i}(K)^\vee[-2].$$

The case $i < 0$ is the vanishing already proved in Theorem 1.2. $\square$

6.1 Summary table

The nonzero cohomology groups of $C_i = \mathbb{Z}_p(i)(\mathcal{O}_K)$ are:

weight i	H^0	H^1	H^2
$i < 0$	0	0	0
$i = 0$	$\mathbb{Z}_p$	$\mathbb{Z}_p$	0
$i = 1$	0	$\widehat{\mathcal{O}_{Kp}^\times}$	0
$i \geq 2$	0	$H^1(K, \mathbb{Z}_p(i))$	$W_{1-i}(K)^\vee$

For $i \geq 2$, noncanonically,

$$H^1(K, \mathbb{Z}_p(i)) \cong \mathbb{Z}_p^{n_K} \oplus W_i(K).$$

6.2 Odd p

Assume p is odd. For $i \geq 2$, Theorems 1.2 and 5.3 give

$$H^2(\mathbb{Z}_p(i)(\mathcal{O}_K)) \cong \begin{cases} 0, & d_K \nmid (1-i), \\ (\mathbb{Z}/p^{c_K+v_p(i-1)}\mathbb{Z})^\vee, & d_K \mid (1-i), \end{cases}$$

and

$$\text{tors } H^1(\mathbb{Z}_p(i)(\mathcal{O}_K)) \cong \begin{cases} 0, & d_K \nmid i, \\ \mathbb{Z}/p^{c_K+v_p(i)}\mathbb{Z}, & d_K \mid i. \end{cases}$$

Here $d_K = [K(\mu_p) : K]$ and $\langle \chi_K(G_K) \rangle = 1 + p^{c_K}\mathbb{Z}_p$.

Example 6.2 (The ring $\mathbb{Z}_p[\zeta_p]$). Assume p is odd. Let ζ_p be a primitive p -th root of unity, put

$$K = \mathbb{Q}_p(\zeta_p), \quad \mathcal{O}_K = \mathbb{Z}_p[\zeta_p],$$

and write

$$C_i^\zeta := \mathbb{Z}_p(i)(\mathbb{Z}_p[\zeta_p]).$$

Then $n_K = p - 1$, and under the identification

$$\mathrm{Gal}(\mathbb{Q}_p(\mu_{p^\infty})/\mathbb{Q}_p) \cong \mathbb{Z}_p^\times$$

the subgroup fixing $K = \mathbb{Q}_p(\mu_p)$ is $1 + p\mathbb{Z}_p$. Hence $H_K = 1 + p\mathbb{Z}_p$, so

$$d_K = 1, \quad c_K = 1.$$

Therefore, for every $m \neq 0$,

$$W_m(K) \cong \mathbb{Z}/p^{1+v_p(m)}\mathbb{Z}.$$

Substituting this into Theorem 1.2 and Corollary 1.3 gives the following noncanonical decompositions in $D(\mathbb{Z}_p)$:

$$\begin{aligned} C_i^\zeta &\simeq 0 & (i < 0), \\ C_0^\zeta &\simeq \mathbb{Z}_p \oplus \mathbb{Z}_p[-1], & C_1^\zeta \simeq (\mathbb{Z}_p^{p-1} \oplus \mathbb{Z}/p\mathbb{Z})[-1], \end{aligned}$$

and, for $i \geq 2$,

$$C_i^\zeta \simeq (\mathbb{Z}_p^{p-1} \oplus \mathbb{Z}/p^{1+v_p(i)}\mathbb{Z})[-1] \oplus (\mathbb{Z}/p^{1+v_p(i-1)}\mathbb{Z})^\vee[-2].$$

Equivalently,

$$H^1(C_i^\zeta) \cong \mathbb{Z}_p^{p-1} \oplus \mathbb{Z}/p^{1+v_p(i)}\mathbb{Z} \quad (i \geq 2), \quad H^2(C_i^\zeta) \cong (\mathbb{Z}/p^{1+v_p(i-1)}\mathbb{Z})^\vee,$$

with no other cohomology in weights $i \geq 2$. For $p = 2$, one has $\mathbb{Z}_2[\zeta_2] = \mathbb{Z}_2$, so the calculation is the case $K = \mathbb{Q}_2$ recorded in Example 5.9.

6.3 $p = 2$

Assume $p = 2$. For $i \geq 2$, Theorems 1.2 and 5.8 give

$$H^2(\mathbb{Z}_2(i)(\mathcal{O}_K)) \cong (\mathbb{Z}/2^{a_1-i(K)}\mathbb{Z})^\vee, \quad \mathrm{tors} H^1(\mathbb{Z}_2(i)(\mathcal{O}_K)) \cong \mathbb{Z}/2^{a_i(K)}\mathbb{Z},$$

where, for $m \neq 0$,

$$a_m(K) = \begin{cases} 1, & s_K = 1 \text{ and } m \text{ is odd,} \\ b_K + v_2(m), & \text{otherwise.} \end{cases}$$

6.4 Mod p shadow

Let

$$\overline{C}_i := C_i \otimes_{\mathbb{Z}_p}^{\mathbb{L}} \mathbb{F}_p.$$

For every r , the universal coefficient sequence gives

$$0 \longrightarrow H^r(C_i)/p \longrightarrow H^r(\overline{C}_i) \longrightarrow H^{r+1}(C_i)[p] \longrightarrow 0.$$

Thus, for $i \geq 2$,

$$H^0(\overline{C}_i) \cong W_i(K)[p],$$

there is a short exact sequence

$$0 \longrightarrow H^1(K, \mathbb{Z}_p(i))/p \longrightarrow H^1(\overline{C}_i) \longrightarrow W_{1-i}(K)^\vee[p] \longrightarrow 0,$$

and

$$H^2(\overline{C}_i) \cong W_{1-i}(K)^\vee/p.$$

When p is odd, $H^0(K, \mathbb{F}_p(m))$ is one-dimensional over $\mathbb{F}_p$ exactly when $d_K \mid m$, and is zero otherwise. Hence the mod p dimensions of the syntomic weights are governed by the cyclotomic period $d_K = [K(\mu_p) : K]$, together with the n_K unit directions coming from $\widehat{\mathcal{O}_{Kp}^\times}/p$.

References

- [1] B. Bhatt and A. Mathew, *Syntomic complexes and p -adic étale Tate twists*, Forum Math. Pi **11** (2023), e1.
- [2] S. Bloch and K. Kato, *p -adic étale cohomology*, Inst. Hautes Etudes Sci. Publ. Math. **63** (1986), 107–152.
- [3] B. Bhatt, M. Morrow, and P. Scholze, *Topological Hochschild homology and integral p -adic Hodge theory*, Publ. Math. IHES **129** (2019), 199–310.
- [4] U. Jannsen, *Continuous étale cohomology*, Math. Ann. **280** (1988), no. 2, 207–245.
- [5] J. S. Milne, *Arithmetic Duality Theorems*, second edition notes, 2006.
- [6] K. Sato, *p -adic étale Tate twists and arithmetic duality*, Ann. Sci. Ec. Norm. Sup. (4) **40** (2007), no. 4, 519–588.
- [7] J.-P. Serre, *Local Fields*, Graduate Texts in Mathematics 67, Springer-Verlag, 1979.
- [8] J.-P. Serre, *Galois Cohomology*, Springer Monographs in Mathematics, Springer-Verlag, 2002.